Assignment #1

Prog. Design and Data Structures

constant < logarithms <
polynomials < exponentials

1.) Take the list of functions and arrange them in ascending order of growth rate.

$n^2 + 5n$, $n\log n$, $1000n$, $\log n$, 2^n, $100n + 5$, $\frac{n^2}{2}$, c

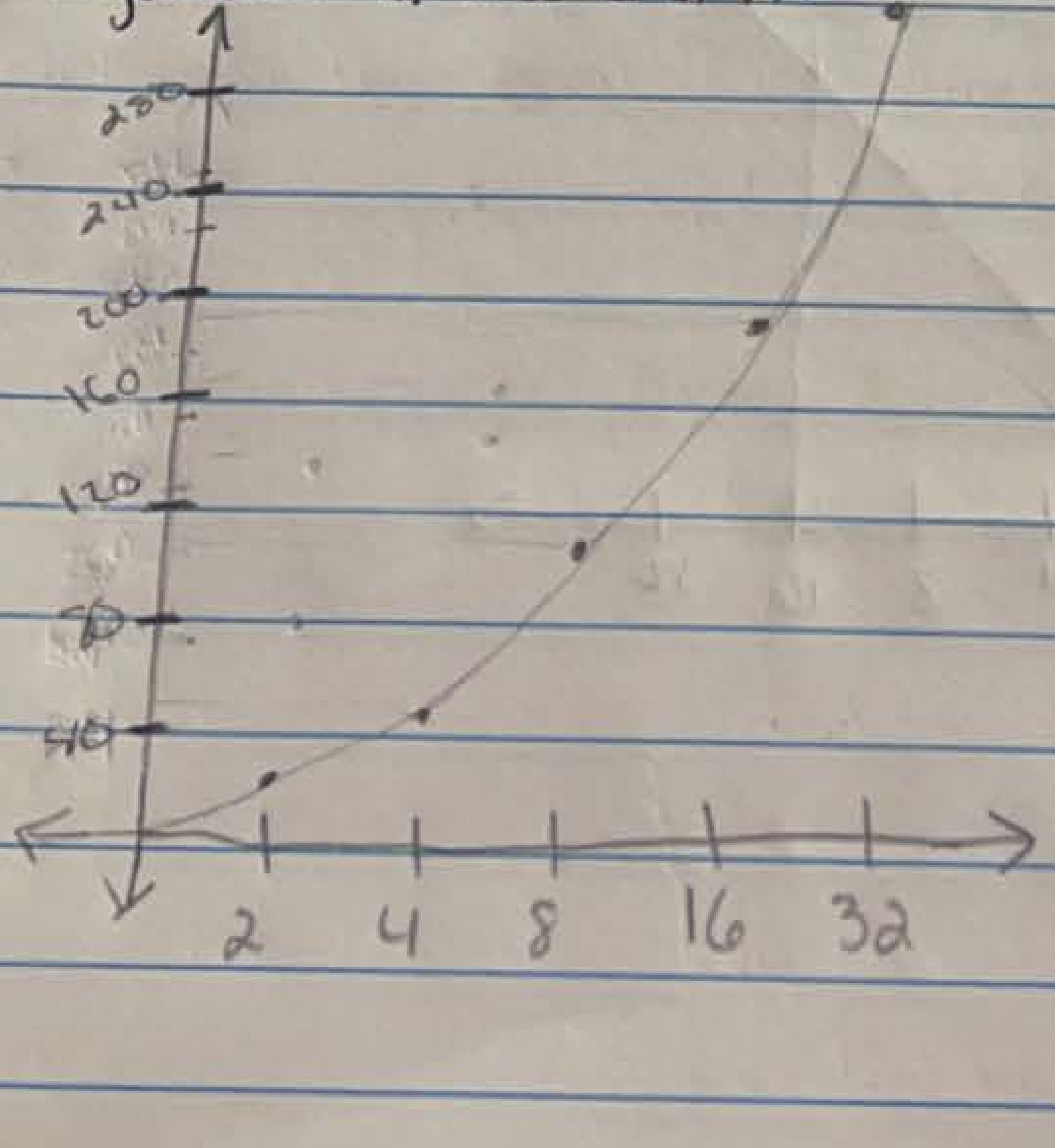
$\{ c, \log n, 100n + 5, 10000n, n\log n, \frac{n^2}{2}, n^2 + 5n, 2^n \}$

2.) What is the worst-case big-O analysis of the follow code fragments?

a.) $O(N)$
b.) $O(N^2)$
c.) n^2

3.) Graph the functions $12n$, $6n\log n$, n^2, n^3, 2^n using a logarithmic scale for the x- and y-axes; that is, if the function value $f(n)$ is y, plot this as a point with x-coordinate at $\log n$ and y-coordinate at $\log y$. Recall that $\log n$ denotes the logarithm in base 2 of n.

$12n$



12(2) = 2
12(4) =
12(8) =
12(16) = 1
12(32) =